HP_power_board

File: HP_power_board.kicad_sch

LP_logic_board

File: LP_logic_board.kicad_sch

pi_power_stub

File: pi_power_stub.kicad_sch

pack_sense_board

File: pack_sense_board.kicad_sch

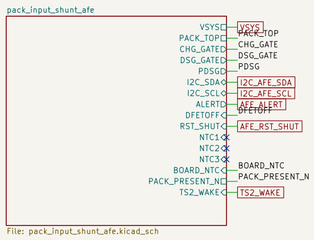
marengo

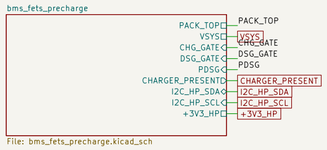
Sheet: /

File: marengo_pdu.kicad_sch

Title: marengo power board rev A

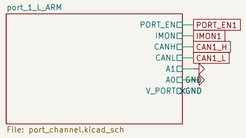
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 1/27

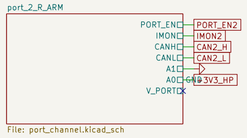


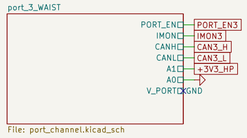


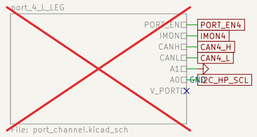


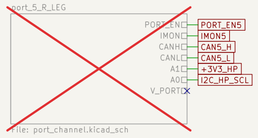


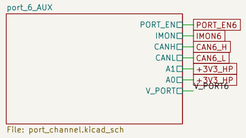




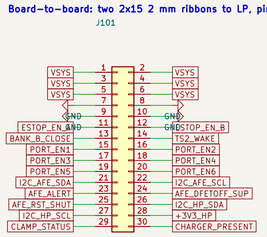


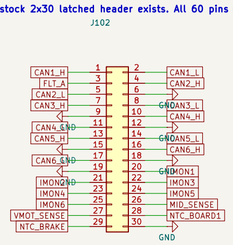


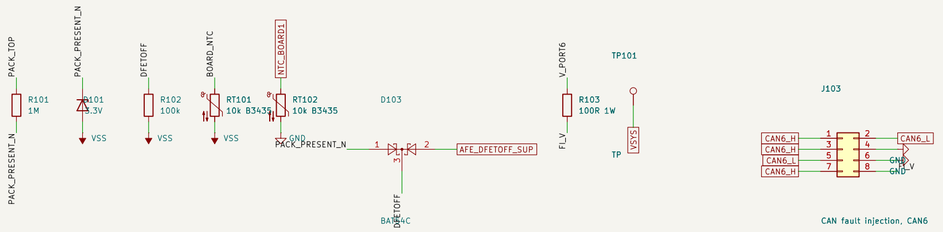




Board-to-board: two 2x15 2 mm ribbons to LP, pin for pin. No stock 2x30 latched header exists. All 60 pins used.







DFETOFF, active high, BOTHOFF (config 0x42): pack sense harness or cell plug pulled, or supervisor backup, opens DSG and CHG. E-stop loop deliberately not wired here: it would deadlock boot (12 V comes after DSG) and drop compute.
Pull-up from PACK_TOP, not LP: it holds with the LP rails down. 54 uA through the presence loop when mated.
Fault injection: test mode, disarmed. Short H-L, H-GND, L-GND, H-VBUS via 100R.

marengo
Sheet: /HP_power_board/
File: HP_power_board.kicad_sch

Title: HP_power_board: high-power board

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 2/27

BAT from the pack + terminal, so the AFE stays up if J2 is pulled
with pack power still connected. Fig 16-2: diode, 100R, 1uF, 100pF. CP1 is 1uF, not 0.47uF: see sheet bms_fets_precharge. Caps 100 V.

PACK_TOP
D1
BAV19W 120V
R15
100R
BAT
C14
1uF 100V
C15
100pF
C16
1uF
VSS
VSS
CP1

J2
2x10 locking header
NTC1
NTC3
PACK_PRESENT_N
B0
B2
B4
B6
B8
B10
B12
B13
B11
B9
B7
B5
B3
B1
NTC2

J2: harness to the pack sense board 2x10, same pin map.
Columns 4-10 zigzag B0..B13: a bent pin shorts 1 cell
across a column, 2 cells along a row.
Pin 3: NTC and presence return, to VSS here, no cell connection.
Pin 4: B0 Kelvin tap, no bond to BAT- (R below). Spare: pin 12.

J1 pack +, J3 pack -: 10 AWG ring-lug pigtails to the XT90 at the pack.

J1
PACK_TOP
M4 terminal, REDCUBE 74651194R
J3
BAT-
M4 terminal, REDCUBE 74651194R
NT1
Net-Tie
VSS
SRP
SRN
RS1
0.3mR CSS4J-4026 5W
GND

Pack negative -> shunt -> system GND. SRP is the cell side,
charge current makes SRP positive (datasheet p5).
AFE VSS ties at BAT-, so VSS sits above GND by I x 0.3 mR
in discharge: 60 mV at the 200 A SCD point.

B0
R24
BAT-
10k

Keeps VC0 defined with the harness out. 9 uA at 60 A.

B13
R1
20R
VC16
C1
220nF
B12
R2
100R
VC12
C2
220nF
B11
R3
100R
VC11
C3
220nF
B10
R4
100R
VC10
C4
220nF
B9
R5
100R
VC9
C5
220nF
B8
R6
100R
VC8
C6
220nF
B7
R7
100R
VC7
C7
220nF
B6
R8
100R
VC6
C8
220nF
B5
R9
100R
VC5
C9
220nF
B4
R10
100R
VC4
C10
220nF
B3
R11
100R
VC3
C11
220nF
B2
R12
100R
VC2
C12
220nF
B1
R13
100R
VC1
C13
220nF
B0
R14
100R
VC0

Cell filter: 100R + 220nF, 20R on VC16 (SLUAA81 2.5, top cell IR drop)
(datasheet p9: RC 20-100R, CC 0.1/0.22/1 uF). 100R gives the balance FET its gate drive.

U1
BQ76952PFBR
BAT
CP1
VC16
VC15
VC14
VC13
VC12
VC11
VC10
VC9
VC8
VC7
VC6
VC5
VC4
VC3
VC2
VC1
VC0
SRP
SRN
VSS
NC
NC
CHG
DSG
PCHG
PDSG
PACK
LD
FUSE
BREG
REGIN
REG1
REG2
REG18
SDA
SCL
ALERT
RST_SHUT
DFETOFF
CFETOFF
HDQ
DCHG
DDSG
TS1
TS2
TS3
CHG_GATE
DSG_GATE
PDSG
PACK_S
LD_S
VSS
C19
2.2uF
VSS
I2C_SDA
I2C_SCL
ALERT
RST_SHUT
DFETOFF
NTC3
BOARD_NTC
NTC1
TS2_F
NTC2

13S per Table 16-3 (p75): cells 1, 2, 16 must be real.
VC0..VC12 = cells 1-12. VC15-VC16 = cell 13.
VC13-15 short to VC12. Cell inputs in use: 1-12 and 16.

SRP
R16
100R
C17
100nF
C18
100pF
SRN
R17
100R
SRP_F
SRN_F

p65, p76: 100R each side, 100nF + 100pF differential, near U1,
symmetric, same PCB side. Kelvin SRP/SRN routed in parallel from RS1.

PACK and LD both sense VSYS, the common port after the FETs
(Fig 16-3 R42-R45). PCHG, FUSE, DCHG, DDSG unused, left floating (p75).

PACK_S
R18
5.1k
R19
5.1k
VSYS
LD_S
R20
5.1k
R21
5.1k
VSYS
RST_SHUT
R22
1k
VSS
TS2_F
R23
10k
TS2_WAKE
NTC1
C20
1nF
VSS
NTC2
C21
1nF
VSS
NTC3
C22
1nF
VSS

Four thermistors: TS1, TS3, CFETOFF (NTC1-3), HDQ (board).
TS2 stays free as SHUTDOWN wake (p22 pin table, section 13):
switch to VSS through 10k (Fig 16-2 R27); 1nF: p65 limit <4nF.
REGIN, BREG tied to VSS, REG1, REG2 open (p75); I2C pull-ups on LP.
CFETOFF as NTC needs DFETOFF set as BOTHOFF to stop CHG by pin (p48).

SIZING (shunt 0.3 mOhm: no 0.25 mOhm 4-terminal part exists: datasheet p22 steps)
OCD1 24 mV = 80 A, 66.7-93.3 A at +/-4 mV, delay 100 ms OCD3 60 A for 1 s (current-based, inside the AFE, the agreed 60 A ceiling)
OCC 12 mV = 40 A, 33.3-48.8 A at -2/+2.65 mV, under the 50 A limit SCD 60 mV = 200 A, 170-230 A, delay 30 us: bank A short protection (105-165 A, about 10 us) acts first
OCD and OCC are set in 2 mV steps. V(OC_ACC) is -2/+2.65 mV below 20 mV, +/-4 mV 20-56 mV, +/-5 mV 56-100 mV. RS1 takes 6.75 W at 150 A: 5 W part, 100 ms path only.

OPEN ON THIS SHEET
1. Balance FETs are captured below (SLUAA81 Fig 2-2). OTP/data memory owed: Cell Balance Min Cell V 3900 mV, Max Cells 4.
2. CP1 raised to 1 uF for the 4+4 FET gate load (p65 rule): worst case 1.69 uC droops 11 V by 1.7 V, pump start about 130 ms.
3. OTP owed: DFETOFF Pin Config 0x42 (active high, BOTHOFF), CFETOFF and HDQ as thermistor inputs, Presence and DFETOFF logic are on HP_power_board.
   Cell 1 interconnect stays 0: VC0 reads the Kelvin B0 tab, VC0 sits below VSS by the 10 AWG return drop, about 90 mV at 60 A (abs max VSS - 0.3 V).
4. RS1 land: CSS4J-4026 on the BVR4026 library footprint, check against the Bourns drawing.

External balance, SLUAA81A Fig 2-2: internal FET current through VC(n-1) input resistor gives Vgs. 3.9 V: 1.59 V min (RCB 46 R). SSM3K56FS is specified at 1.5 V for 50 mA only and 1.8 V for 200 mA: check Rds at 150 mA on the bench, top cell worst case 2.35 V.
Rbal 27R: 140-150 mA external at 3.9-4.2 V, plus 17-19 mA internal. Cell Balance Max Cells 4: about 2.5 W total.

Q1
R25
27R 1W 2512
BG1
R26
1k
D2
BZT52C5V1
SSM3K56FS
Q2
R27
27R 1W 2512
BG2
R28
1k
D3
BZT52C5V1
SSM3K56FS
Q3
R29
27R 1W 2512
BG3
R30
1k
D4
BZT52C5V1
SSM3K56FS
Q4
R31
27R 1W 2512
BG4
R32
1k
D5
BZT52C5V1
SSM3K56FS
Q5
R33
27R 1W 2512
BG5
R34
1k
D6
BZT52C5V1
SSM3K56FS
Q6
R35
27R 1W 2512
BG6
R36
1k
D7
BZT52C5V1
SSM3K56FS
Q7
R37
27R 1W 2512
BG7
R38
1k
D8
BZT52C5V1
SSM3K56FS
Q8
R39
27R 1W 2512
BG8
R40
1k
D9
BZT52C5V1
SSM3K56FS
Q9
R41
27R 1W 2512
BG9
R42
1k
D10
BZT52C5V1
SSM3K56FS
Q10
R43
27R 1W 2512
BG10
R44
1k
D11
BZT52C5V1
SSM3K56FS
Q11
R45
27R 1W 2512
BG11
R46
1k
D12
BZT52C5V1
SSM3K56FS
Q12
R47
27R 1W 2512
BG12
R48
1k
D13
BZT52C5V1
SSM3K56FS
Q13
R49
27R 1W 2512
BG13
R50
1k
D14
BZT52C5V1
SSM3K56FS

Pack terminals, sense header, shunt, BQ76952 AFE. Part numbers verify. TI SLUSE13B.

marengo

Sheet: /HP_power_board/pack_input_shunt_afe/
File: pack_input_shunt_afe.kicad_sch

Title: pack_input_shunt_afe

Size: A2 | Date: 2026-11-06 | **Rev: A-wip**
KiCad E.D.A. 10.0.8 | Id: 3/27

PDSG FET bypasses DSG only (Fig 16-1): source at CD. Driver sinks 30 uA, so gate-source >= 1M (p45).

R210 1M
Q209 DMP10H400SE
PDSG
PDSG_R

R209
A
B
TAB
PDSG_R
DSG_OUT
100R PWR263S-35-1000F

Common drain (CD): Fig 16-3 pin numbering, gates referenced to the outer sources. Body diodes oppose each other.

CD
Q201 Q202 Q203 Q204 Q205 Q206 Q207 Q208
R201 R202 R203 R204 R205 R206 R207 R208 4.7R
CHG_G DSG_G
IPT015N10N5
PACK_TOP
DSG_OUT

CHG bank: sources on the stack

DSG bank: sources on the output

R5201 0.3mR CSS4J-4026 5W
VSYS_SENS_P VSYS_SENS_N
TP201 VSYS
D207 SMDJ58A GND
C207 100nF 100V
C208 100nF 100V 100uF 80V HHXF800GRA101MJH0G
C209 GND

CHG drive, Fig 16-3 R37/R31/D5.

CHG_GATED
R211 5.1k
R212 10M
D201 MMSZ5246B 16V
CHG_G
PACK_TOP

Fig 16-3 C28 C29: across the FETs, two in series.
C201 100nF 100V
C202 100nF 100V

Stack spike: 3 x 100 uF hybrid (right, below U201) take the 0.36 J at 2 uH, 600 A LC peak 77 V at -20% C. TVS is a backstop (conducts from 64.4 V).
D206 SMDJ58A GND

Fig 16-3 Q4 D7 R33 R35 D8: If the output is yanked below GND, Q210 holds the DSG gate at its source.

DSG drive. On: 10.2k (Fig 16-3 R39+R41). Off: PNP Q211 local turn-off, base through 220R + diode to the DSG pin. Not the 7.5k of Fig 16-3 (see note 2).

DSG_GATED
R213 5.1k R214 5.1k
D202 BAV19W R215 220R
DSG_OFF_B
R216 10M D203 MMSZ5246B 16V
R218 10k D204 MMSZ5246B 16V
D205 BAV19W GND
R217 10M
Q210 BSS123
Q211 MMBT2907A
DSG_G

U201 INA228AIDGSR
VSYS
+3V3_HP
VSYS_SENS_P VSYS_SENS_N
C206 100nF GND
Vbus Vin+ Vin- VS GND A1 A0 SDA SCL Alert
I2C_HP_SCL I2C_HP_SDA I2C_HP_SCL
Main path meter, PLAN 3.9. Address 0x4C: A1 to SCL, A0 to GND. +/-40.96 mV range = +/-136.5 A. Bank A short sense reads this shu

C203 C204 C205 100uF 80V HHXF800GRA101MJH0G

Charge input, common port: the charger joins on VSYS, DSG body diode side, as the BQ76952 expects (PACK pin sees it, p45). Also wakes the AFE from ship mode through LD.
PLAN 3.7 says reverse-polarity FET. With a reversed charger the blocker sees 54.6 + 54.6 = 109 V: above the 100 V FET and the LM74700 (75 V, SNOSD17G). So a 150 V Schottky: Vishay V10PM15, pads 1, 2 anode, tab K, 0.6 V, 2.4 W at 4 A. SEE RATIONALE.log
CHARGER_PRESENT: CHG_IN only, so VSYS cannot fake it: high above 34 V, 3.2 V at 54.6 V.

SIZING (design/README 2.2)
Conduction, 80 A: 2 x 0.6 mOhm hot x 80^2 = 7.7 W total. At 10 to 30 A: 0.1 to 1.1 W for these two banks.
FET: IPT015N10N5 Rds 1.5 mOhm max at Vgs 10 V, Vgs +/-20 V, so 11 V drive and 16 V zeners are in range (p69 needs >= 15 V).
Shunt voltage rule p69: 13S x 5 V = 65 V DC, 130 V peak. 100 V FETs meet DC. Peak: both TVS parts hold it under 100 V.

J201 al, charger +, 74651194R
J202 al, charger -, 74651194R
CHG_IN
D208 SMDJ58CA GND
D209 BAV19W
R219 1M
CHARGER_PRESENT
R220 62k
D210 MMSZ4685 3.6V
C210 1nF
GND

D220 V10PM15-M3/I 150V 10A
A1 A2 K
CHG_IN CHG_IN VSYS

SOA must cover 600 A prospective for about 50 us. Reworked after gate 3:
1. Prospective current: 600 A / 4 = 150 A per FET. On-state part: 0.6 mOhm x 600 A = 0.36 V, 216 W bank, nothing.
2. Turn-off: TI gives only 2 us typ, 20 us max at 20 nF for the DSG pin, pulsed, stopping at LD (section 16.5). Ours is 48 nF, so a bare 220R is unsupported.
   Q211: 25 mA base through 220R, 600 mA collector, all gate charge 844 nC max in about 1.5 us. 10 us SOA at 54 V is about 300 A per FET: PASSES at 150 A, with sharing unmodelled: scope at bring-up.
3. Spike: p72 warns fast turn-off raises the top of stack. 3 x 100 uF hybrid at top right take 0.36 J: LC peak 73 V, 77 V at -20% C. FET stays under 100 V without avalanche.

CP1 is 1 uF on sheet 1: 8 x 211 nC max = 1.69 uC at p65's rule, droop 1.7 V from 11 V.
PDSG not PCHG: p45 PDSG charges the LOAD before DSG closes; PCHG charges a LOW BATTERY. README 2.1 described the PDSG job.
100R: peak 54.6^2/100 = 29.8 W. VSYS load 1.46 mF (hold-up via Q1101 body diode, plus VSYS bulk): 2.2 J per power-up, tau 0.146 s. PWR263S chart: 10 J at 100 ms.
Set OTP: PREDISCHARGE on, Prediscarge Timeout 0 (exit on voltage only), Stop Delta 2500 mV: a timeout closes DSG whatever the voltage.

Fig 16-1, Fig 16-3 of TI SLUSE13B. High side, common drain, common port.

marengo

Sheet: /HP_power_board/bms_fets_precharge/
File: bms_fets_precharge.kicad_sch

Title: bms_fets_precharge: BMS CHG/DSG banks, prediscarge, charge input

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 4/27

U301

TPS48100QDGXRQ1

U302

TPS48100QDGXRQ1

TP301

TP302

Bank A: VSYS to MID

Drain toward VSYS, source toward VMOT: body diodes conduct VMOT to VSYS, so regen still reaches the pack with the banks open (PLAN 3.4).

Q301 Q303 Q305

IPT015N10N5

Bank B: MID to VMOT

Q302 Q304 Q306

IPT015N10N5

VMOT precharge (IPT015N10N5 SOA is 0.7 A at 54 V, 10 ms, so a hard close on 1.8 mF is out). U302 EN and INP2 on ESTOP_EN_B: loop closed and latch armed, G2 precharges at once, bank B stays open.
BANK_B_CLOSE (INP1, 100k down) closes bank B on command once VMOT is within 2 V of VSYS. Reaching 2 V takes 0.595 s, 0.71 s at +20% C: firmware timeout 0.85 s. EN low kills G1 and G2. Energy 2.68 J, tau 180 ms, peak 29.8 W.

Status (TPS48100 has no VDS output): MID, VMOT dividers 1M / 56k, 3.3 V = 62 V, plus VSYS on LP. Bank A short reads MID near VSYS with both open and VMOT low.

Bank A short protection: ISCP 10.5k = 40 mV (31.6–49.5 mV) on RS201 0.3 mOhm = 105–165 A, under BMS SCD and over OCD1 max 93 A. About 10 us, latch-off (100k + 680 pF across TMR), FLT_A to t

Q307

IPT015N10N5

R318

TP303

100R PWR263S-35-1000F

TP304 TP305

TPS48100-Q1 simple-driver mode (SLVSGC0 Fig 7-11). No firmware in the enable path.

marengo

Sheet: /HP_power_board/estop_banks/
File: estop_banks.kicad_sch

Title: estop_banks: E-stop banks A and B, VMOT precharge

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 5/27

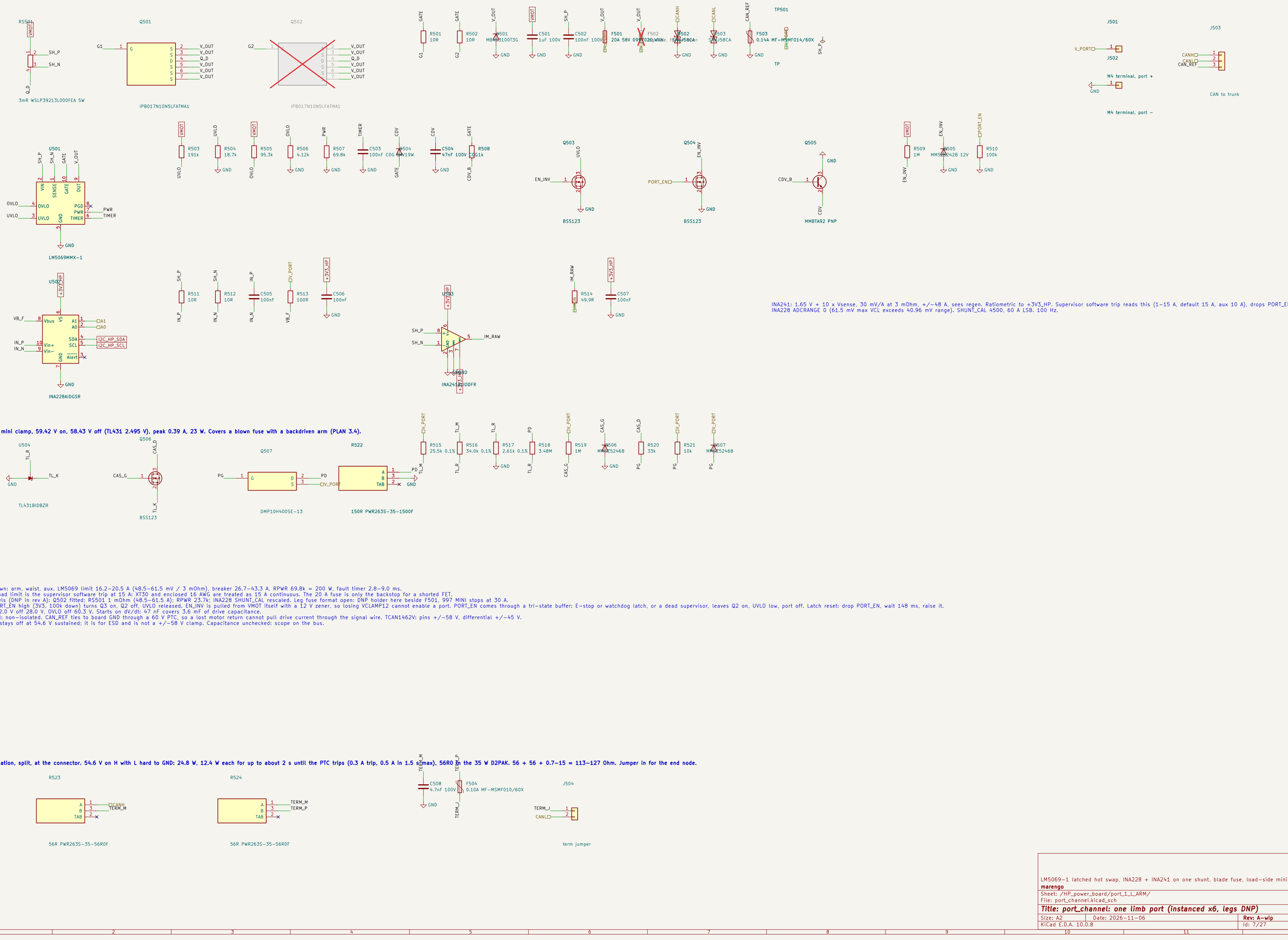
Bulk: 18 x 100 uF 80 V hybrid, 1.8 mF. 220 uF 80 V 10x12.5 does not exist. Can is 10 x 16.5 mm: footprint is the closest library match, real land owed.

C401 100uF 80V HHXF800GRA101MJH0G
C402 100uF 80V HHXF800GRA101MJH0G
C403 100uF 80V HHXF800GRA101MJH0G
C404 100uF 80V HHXF800GRA101MJH0G
C405 100uF 80V HHXF800GRA101MJH0G
C406 100uF 80V HHXF800GRA101MJH0G
C407 100uF 80V HHXF800GRA101MJH0G
C408 100uF 80V HHXF800GRA101MJH0G
C409 100uF 80V HHXF800GRA101MJH0G
C410 100uF 80V HHXF800GRA101MJH0G
C411 100uF 80V HHXF800GRA101MJH0G
C412 100uF 80V HHXF800GRA101MJH0G
C413 100uF 80V HHXF800GRA101MJH0G
C414 100uF 80V HHXF800GRA101MJH0G
C415 100uF 80V HHXF800GRA101MJH0G
C416 100uF 80V HHXF800GRA101MJH0G
C417 100uF 80V HHXF800GRA101MJH0G
C418 100uF 80V HHXF800GRA101MJH0G
D401 SMDJ58A
D402 SMDJ58A
C419 1uF 100V
C420 4.7uF 25V
R401 464k
R402 49.9k
R403 9.09k
U403 LM4040AIM3-2.5
U401 TPS7A4001DGNR
R404 499k 0.1%
R405 604k 0.1%
R406 51.1k 0.1%
R407 19.1M
C421 100pF C0G
C422 100nF
Q402 BSS123
J401
J402 M4 terminal, Arcol HS25 10R +
J403 brake R NTC
M4 terminal, Arcol HS25 10R −
U402 TLV1805DBVR
C423 1uF 25V
R408 4.7R
R409 100k
U404 UCC27517DBVR
Q401 IPT015N10N5
RS401 2mR CSS4J-4026K-2L00F 6W
D403 MBRB30H100CT-E3/81
R410 10R
R411 10R
C424 100nF
R412 100R
C425 100nF
U405 INA228AIDGSR

Clamp: 56.77 V on, 55.77 V off (VOH 11.7 V, 19.1M hysteresis). Spread 56.08–57.46 V full temperature. Arcol HS25 10 Ohm 25 W clad, external on 2 mm plate: 322 W, 5.7 A.
FET, 2 mOhm shunt (INA228 SHUNT_CAL 2000, ADCRANGE 0), 30 A Schottky and M4 terminals are sized for the leg stage 2 Ohm resistor, 29 A, 1.7 kW short time: only the external resistor swaps.
INA228 0x4D: clamp power and energy on its own accumulator. Budget about 5 mA, 0.23 W in the LDO.
Failure: stuck-on 322 W: NC thermostat on the HS25 plate sits in the E-stop loop, so it opens the banks in hardware, supervisor hung or not. NTC warns first. SMDJ58A stays off under 64.4 V, spike protection only.
Regen decision (coordinator 11-06): 10 Ohm in rev A. 30 A regen against 5.75 A clamp: BMS opening mid-brake rises 13.5 V/ms, drives stop at 60 V in about 185 us, torque drops. Leg stage fits 2 Ohm.

Regen clamp powers itself from VMOT, works with supervisor, BMS and pack all off (PLAN 3.4).

marengo	
Sheet: /HP_power_board/vmot_bulk_clamp/	
File: vmot_bulk_clamp.kicad_sch	
Title: vmot_bulk_clamp: VMOT bulk, TVS, self-powered regen clamp	
Size: A2	Date: 2026-11-06
KiCad E.D.A. 10.0.8	Rev: A-wip
	Id: 6/27



INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RSS01 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113–127 Ohm. Jumper in for the end node.

LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

marengo

Sheet: /HP_power_board/port_1_L_ARM/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 7/27

RS601
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W

Q601
G1
G
S
S
S
D
S
S
S
V_OUT
V_OUT
Q_D
V_OUT
V_OUT
V_OUT
IPB017N10N5LFATMA1

Q602
G2
V_OUT
V_OUT
Q_D
V_OUT
V_OUT
V_OUT
IPB017N10N5LFATMA1

GATE
R601
10R
G1
GATE
R602
10R
G2
V_OUT
D601
MBRS5100T3G
GND
VMOT
C601
1uF 100V
GND
SH_P
C602
100nF 100V
GND
V_OUT
F601
20A 58V 0977020.WXN
V_OUT
F602
Do not use, for 58V CAN
CANH
SMAJ58CA
GND
CANL
SMAJ58CA
GND
CAN_REF
F603
0.14A MF-MSMF014/60X
GND
TP601
TP
SH_P

J601
V_PORT
1
J602
M4 terminal, port +
GND
1
M4 terminal, port −

J603
CANH
CANL
CAN_REF
1
2
3
CAN to trunk

U601
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
OVLO
UVLO
UVLO
GND
PGD
PWR
TIMER
PWR
TIMER
GND
LM5069MMX-1

VMOT
R603
191k
UVLO
UVLO
R604
18.7k
GND
VMOT
R605
95.3k
OVLO
OVLO
R606
4.12k
GND
PWR
R607
69.8k
GND
TIMER
C603
100nF C0G 1kV
GND
CDV
D604
5V19W
GATE
CDV
C604
47nF 100V
GND
GATE
R608
C0G1k
CDV_B

Q603
UVLO
EN_INV
1
2
3
GND
BSS123

Q604
EN_INV
PORT_EN
1
2
3
GND
BSS123

Q605
GND
CDV_B
1
2
3
CDV
MMBTA92 PNP

VMOT
R609
1M
EN_INV
D605
MMSZ5242B 12V
GND
EN_INV
PORT_EN
R610
100k
GND

U602
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
IN_P
IN_N
Vin+
Vin−
GND
Alert
A1
A0
I2C_HP_SDA
I2C_HP_SCL
GND
INA228AIDGSR

SH_P
R611
10R
IN_P
SH_N
R612
10R
IN_N
IN_P
C605
100nF
IN_N
CV_PORT
R613
100R
VB_F
+3V3_HP
C606
100nF
GND

U603
+3V3_HP
SH_P
SH_N
+IN
−IN
VS
GND
OUT
IM_RAW
C609
GND
+3V3_HP
INA241B1IDDFR

IM_RAW
R614
49.9R
IM_RAW
+3V3_HP
C607
100nF
GND

INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

U604
TL_R
GND
TL_K
TL431BIDBZR

Q606
CAS_D
CAS_G
1
3
2
TL_K
BSS123

Q607
PG
G
D
S
1
2
3
PD
V_PORT
DMP10H400SE-13

R622
A
B
TAB
1
3
2
PD
GND
150R PWR263S-35-150F

V_PORT
R615
25.5k 0.1%
TL_M
TL_M
R616
34.0k 0.1%
TL_R
TL_R
R617
2.61k 0.1%
GND
PD
R618
3.48M
TL_R
V_PORT
R619
1M
CAS_G
CAS_G
Q606
MMBZ5246B
GND
CAS_D
R620
33k
PG
V_PORT
R621
10k
PG
V_PORT
Q607
MMBZ5246B
PG

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7–15 = 113–127 Ohm. Jumper in for the end node.

R623
A
B
TAB
1
3
2
CANH
TERM_M
56R PWR263S-35-56R0F

R624
A
B
TAB
1
3
2
TERM_M
TERM_P
56R PWR263S-35-56R0F

TERM_M
C608
4.7nF 100V
GND
TERM_P
F604
0.10A MF-MSMF010/60X
TERM_J

J604
TERM_J
CANL
1
2
term jumper

LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.
marengo
Sheet: /HP_power_board/port_2_R_ARM/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2 | Date: 2026-11-06 | **Rev: A-wip**
KiCad E.D.A. 10.0.8 | Id: 8/27

RS701
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W

Q701
G1
G
S
S
S
D
D
D
V_OUT
V_OUT
Q_D
V_OUT
V_OUT
V_OUT
IPB017N10N5LFATMA1

Q702
G2
V_OUT
V_OUT
Q_D
V_OUT
V_OUT
V_OUT
IPB017N10N5LFATMA1

GATE
R701
10R
G1
GND

GATE
R702
10R
G2
GND

V_OUT
D701
MBRB2100T3G
GND

VMOT
C701
1uF 100V
GND

SH_P
C702
100nF 100V
GND

V_OUT
F701
20A 58V 0R7020.WXN
V_OUT
F702
Do not place, for SMAJ58CA
CANH
D702
SMAJ58CA
GND

CANL
D703
SMAJ58CA
GND

CAN_REF
F703
0.14A MF-MSMF014/60X
GND

TP701
TP
SH_P

J701
V_PORT
1
J702
M4 terminal, port +
GND
1
M4 terminal, port –

J703
CANH
CANL
CAN_REF
1
2
3
CAN to trunk

U701
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
OVLO
UVLO
UVLO
GND
PGD
PWR
TIMER
PWR
TIMER
GND
LM5069MMX-1

VMOT
R703
191k
UVLO
UVLO
R704
18.7k
GND
VMOT
R705
95.3k
OVLO
OVLO
R706
4.12k
GND
PWR
R707
69.8k
GND
TIMER
C703
100nF C0G
GND
CDV
D704
5V19W
GATE
CDV
C704
47nF 100V
GND
GATE
R708
C0G1k
CDV_B

Q703
UVLO
EN_INV
1
2
3
GND
BSS123

Q704
EN_INV
PORT_END
1
2
3
GND
BSS123

Q705
GND
CDV_B
1
2
3
CDV
MMBTA92 PNP

VMOT
R709
1M
EN_INV
EN_INV
D705
MMSZ5242B 12V
GND
PORT_EN
R710
100k
GND

U702
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
IN_P
IN_N
Vin+
Vin-
GND
Alert
A1
A0
I2C_HP_SDA
I2C_HP_SCL
GND
INA228AIDGSR

SH_P
R711
10R
IN_P
SH_N
R712
10R
IN_N
IN_P
C705
100nF
IN_N
CV_PORT
R713
100R
VB_F
+3V3_HP
C706
100nF
GND

U703
+3V3_HP
SH_P
SH_N
+ IN
- IN
Vs
Rs
OUT
GND
IM_RAW
+3V3_HP
INA241A3IDDFR

IM_RAW
R714
49.9R
IM_RAW
+3V3_HP
C707
100nF
GND

INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/-48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

U704
TL_R
GND
TL_K
TL431BIDBZR

Q706
CAS_D
CAS_G
1
2
3
TL_K
BSS123

Q707
PG
G
D
S
1
2
3
PD
V_PORT
DMP10H400SE-13

R722
A
B
TAB
1
3
2
PD
GND
150R PWR263S-35-150F

V_PORT
R715
25.5k 0.1%
TL_M

TL_M
R716
34.0k 0.1%
TL_R

TL_R
R717
2.61k 0.1%
GND

PD
R718
3.48M
TL_R

V_PORT
R719
1M
CAS_G

CAS_G
D706
MMSZ5246B
GND

CAS_D
R720
33k
PG

V_PORT
R721
10k
PG

V_PORT
D707
MMSZ5246B
PG

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/-58 V, differential +/-45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/-58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7–15 = 113–127 Ohm. Jumper in for the end node.

R723
A
B
TAB
1
3
2
CANH
TERM_M
56R PWR263S-35-56R0F

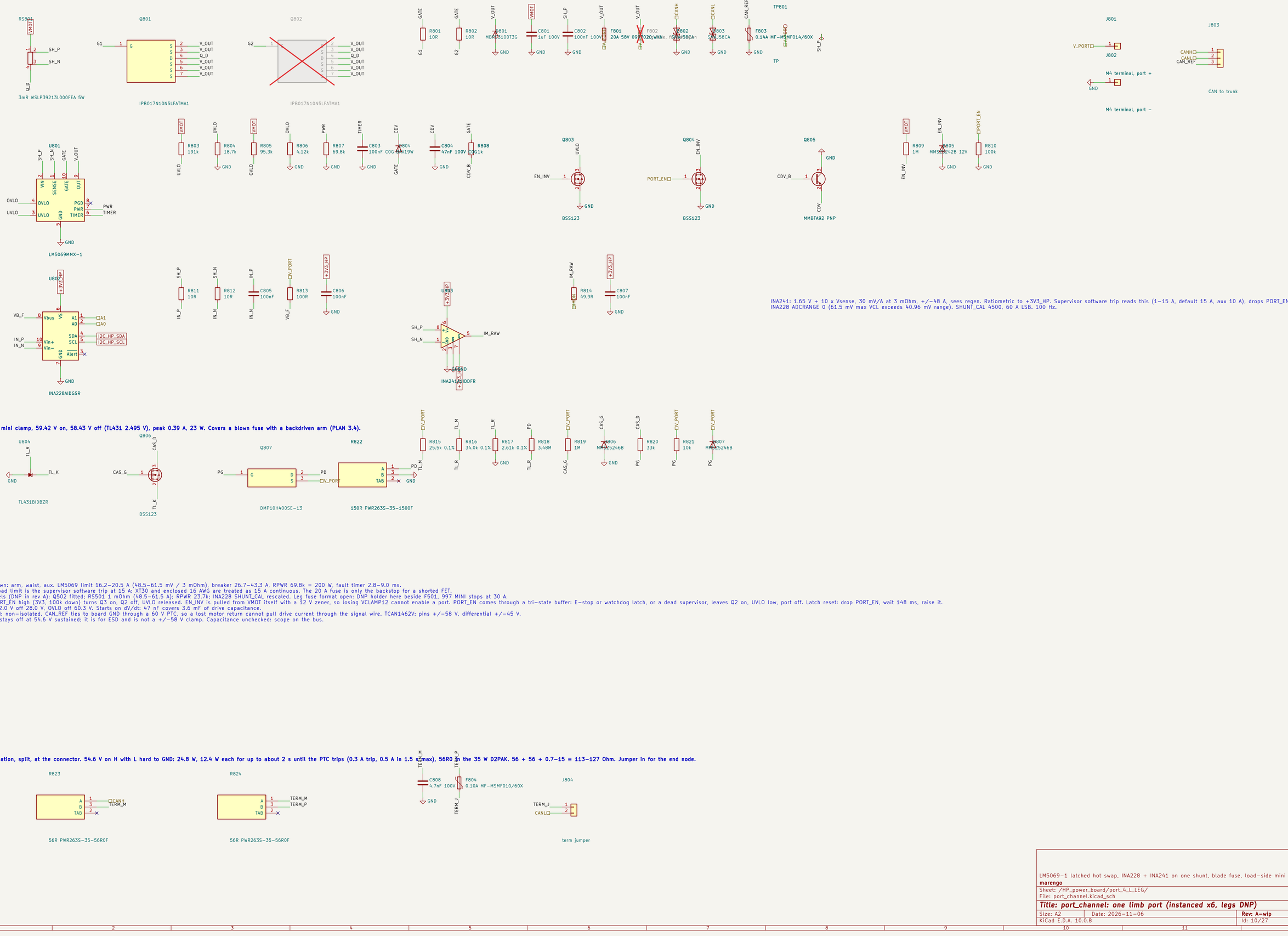
R724
A
B
TAB
1
3
2
TERM_M
TERM_P
56R PWR263S-35-56R0F

TERM_M
C708
4.7nF 100V
GND

TERM_P
F704
0.10A MF-MSMF010/60X
TERM_J

J704
TERM_J
CANL
1
2
term jumper

LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.
marengo
Sheet: /HP_power_board/port_3_WAIST/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2 | Date: 2026-11-06 | **Rev: A-wip**
KiCad E.D.A. 10.0.8 | Id: 9/27

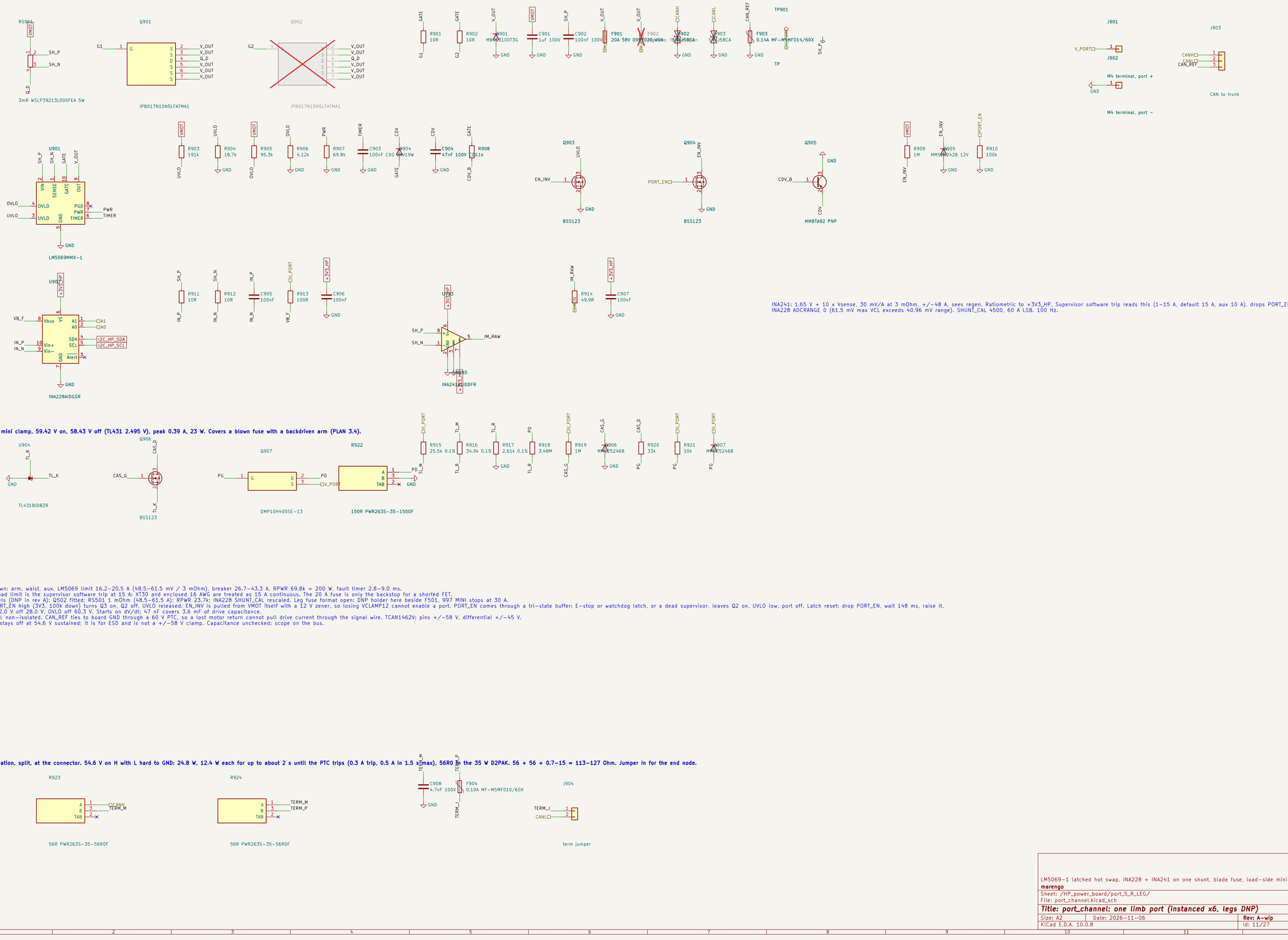


Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.

LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.	
marengo	
Sheet: /HP_power_board/port_4_L_LEG/	
File: port_channel.kicad_sch	
Title: port_channel: one limb port (instanced x6, legs DNP)	
Size: A2	Date: 2026-11-06 — **Rev: A-wip**
KiCad E.D.A. 10.0.8	Id: 10/27


RS901
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q901
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q902
G2
IPB017N10N5LFATMA1
GATE
R901 10R
R902 10R
V_OUT
D901 MBRB2100T3G
C901 1uF 100V
C902 100nF 100V
SH_P
F901 20A 58V 09PT020.WXN
F902 Do Not Populate, for Testing Only
CANH
SMAJ58CA
CANL
CAN_REF
F903 0.14A MF-MSMF014/60X
TP901
TP
SH_P
J901
V_PORT
J902
M4 terminal, port +
GND
M4 terminal, port −
J903
CANH
CANL
CAN_REF
CAN to trunk
U901
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
UVLO
GND
PGD
PWR
TIMER
LM5069MMX-1
R903 191k
R904 18.7k
R905 95.3k
R906 4.12k
R907 69.8k
C903 100nF C0G
D904 4V19W
C904 47nF 100V
R908 C0G1k
CDV
CDV_B
Q903
EN_INV
BSS123
Q904
PORT_END
Q905
MMBTA92 PNP
R909 1M
D905 MMSZ5242B 12V
R910 100k
PORT_EN
U902
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
Alert
I2C_HP_SDA
I2C_HP_SCL
IN_P
IN_N
Vin+
Vin−
INA228AIDGSR
R911 10R
R912 10R
C905 100nF
R913 100R
C906 100nF
U903
IM_RAW
INA241A3IDDFR
R914 49.9R
C907 100nF
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load−side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U904
TL_R
TL_K
TL431BIDBZR
Q906
CAS_D
CAS_G
BSS123
Q907
PG
PD
DMP10H400SE-13
R922
150R PWR263S-35-150F
R915 25.5k 0.1%
R916 34.0k 0.1%
R917 2.61k 0.1%
R918 3.48M
R919 1M
R906 MMSZ5246B
R920 33k
R921 10k
R907 MMSZ5246B
Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri−state buffer: E−stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non−isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.
R923
CANH
TERM_M
56R PWR263S-35-56R0F
R924
TERM_P
TERM_M
TERM_P
C908 4.7nF 100V
F904 0.10A MF-MSMF010/60X
J904
TERM_J
CANL
term jumper
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_5_R_LEG/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 11/27

RS1001
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W

Q1001
G1
V_OUT
Q_D
IPB017N10N5LFATMA1

Q1002
G2
IPB017N10N5LFATMA1

GATE
R1001 10R
G1
GATE
R1002 10R
G2
V_OUT
D1001 MBRB2100T3G
GND
VMOT
C1001 1uF 100V
SH_P
C1002 100nF 100V
V_OUT
F1001 20A 58V 09P1020.WXN
F1002 Do not populate, for testing only
CANH
D1002 SMAJ58CA
CANL
D1003 SMAJ58CA
CAN_REF
F1003 0.14A MF-MSMF014/60X
TP1001
TP
SH_P

J1001
V_PORT
J1002
M4 terminal, port +
GND
M4 terminal, port −

J1003
CANH
CANL
CAN_REF
CAN to trunk

U1001
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
UVLO
GND
PGD
PWR
TIMER
LM5069MMX-1

VMOT
R1003 191k
UVLO
UVLO
R1004 18.7k
GND
VMOT
R1005 95.3k
OVLO
OVLO
R1006 4.12k
GND
PWR
R1007 69.8k
GND
TIMER
C1003 100nF C0G
GND
CDV
D1004 5V19W
GATE
CDV
C1004 47nF 100V
GND
GATE
R1008 C0G1k
CDV_B

Q1003
UVLO
EN_INV
BSS123
GND

Q1004
EN_INV
PORT_EN
BSS123
GND

Q1005
GND
CDV_B
MMBTA92 PNP
CDV

VMOT
R1009 1M
EN_INV
EN_INV
D1005 MMSZ5242B 12V
GND
PORT_EN
R1010 100k
GND

U1002
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
IN_P
IN_N
Vin+
Vin−
GND
Alert
A1
A0
I2C_HP_SDA
I2C_HP_SCL
INA228AIDGSR

SH_P
R1011 10R
IN_P
SH_N
R1012 10R
IN_N
IN_P
C1005 100nF
IN_N
CV_PORT
R1013 100R
VB_F
+3V3_HP
C1006 100nF
GND

U1003
+3V3_HP
SH_P
SH_N
IN+
IN−
VS
GND
IM_RAW
INA241B1IDDFR
GND

IM_RAW
R1014 49.9R
IM_RAW
+3V3_HP
C1007 100nF
GND

INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN. INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load−side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

U1004
TL_R
GND
TL_K
TL431BIDBZR

Q1006
CAS_D
CAS_G
TL_K
BSS123

Q1007
PG
G
D
S
PD
V_PORT
DMP10H400SE-13

R1022
A
B
TAB
PD
GND
150R PWR263S-35-150F

V_PORT
R1015 25.5k 0.1%
TL_M
TL_M
R1016 34.0k 0.1%
TL_R
TL_R
R1017 2.61k 0.1%
GND
PD
R1018 3.48M
TL_R
V_PORT
R1019 1M
CAS_G
CAS_G
D1006 MMSZ5246B
GND
CAS_D
R1020 33k
PG
V_PORT
R1021 10k
PG
V_PORT
D1007 MMSZ5246B
PG

Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, so losing VCLAMP12 cannot enable a port. PORT_EN comes through a tri−state buffer: E−stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non−isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.

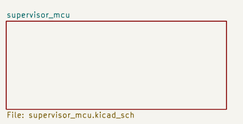
R1023
A
B
TAB
CANH
TERM_M
56R PWR263S-35-56R0F

R1024
A
B
TAB
TERM_M
TERM_P
56R PWR263S-35-56R0F

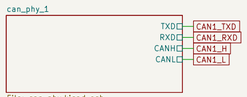
TERM_M
C1008 4.7nF 100V
GND
TERM_P
F1004 0.10A MF-MSMF010/60X
TERM_J

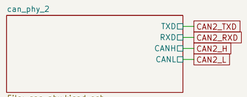
J1004
TERM_J
CANL
term jumper

LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_6_AUX/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2 | Date: 2026−11−06 | **Rev: A−wip**
KiCad E.D.A. 10.0.8 | Id: 12/27

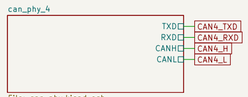






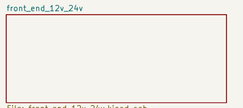


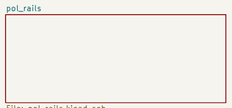


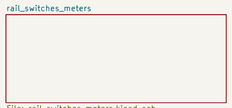


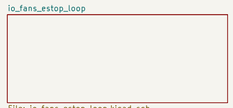




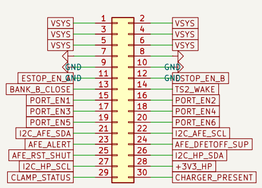


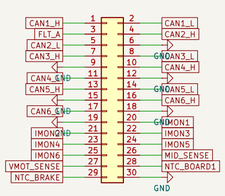




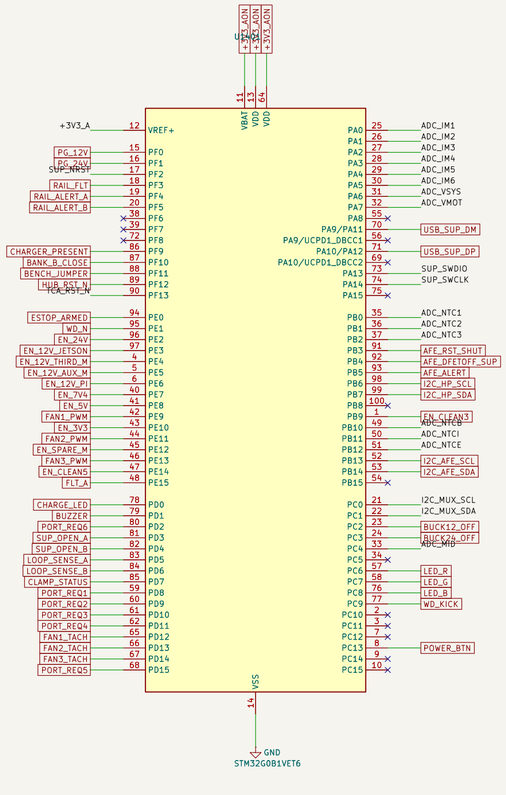


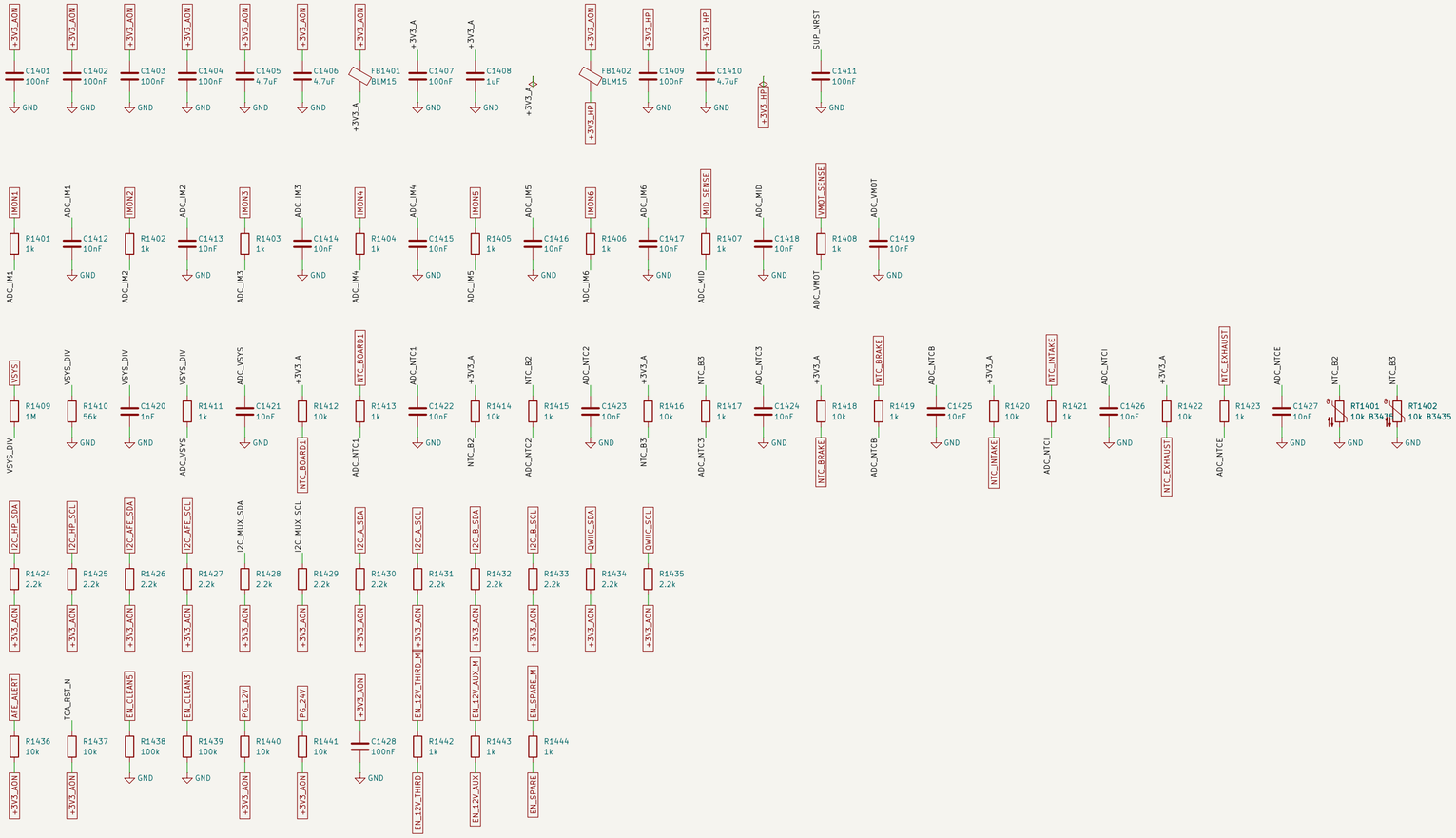
Board-to-board: mirror of HP J101, J102.





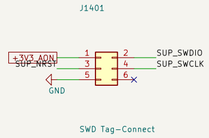
marengo		
Sheet: /LP_logic_board/		
File: LP_logic_board.kicad_sch		
Title: LP_logic_board: low-power board		
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 13/27

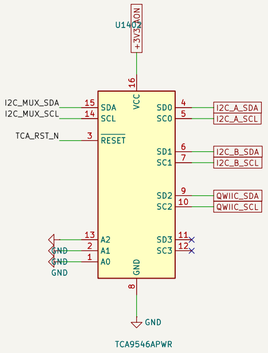




G0B1VET6 pin plan: parts check 68 of 68 against ST pin data. Extra lines beyond the plan: PC2/PC3 buck off, PF0/PF1 PG, PF3–5 rail flags, PB9 EN_CLEAN3. PF10 was BOARD_PRESENT, now BANK_B_CLOSE. PC9 WD_KICK, PE15 FLT_A, PE0 ESTOP_ARMED, PE1 WD_N. PD8–11, PD15, PD2 are now PORT_REQ, gated in hardware to PORT_EN.
I2C: 3 hardware buses. I2C1 HP meters. I2C2 AFE. I2C3 to TCA9546A at 0x70 (ch0 meters A, ch1 meters B, ch2 Qwiic, ch3 spare).
Five NTC channels sit on 10k to the ADC reference. NTC_BOARD1 also reads open when the HP board is missing: replaces BOARD_PRESENT.

12 V LOAD SHED (budget sums to 22 A on a 12.5 A converter, DC limit 14.2 A min). U1321 meters the real total on +12V: amps are not summed across rails. Start values, tune on the bench.
1. Above 10.5 A, 500 ms: fans to reduced duty (FANx_PWM).
2. Above 11.0 A, 500 ms: third board off (EN_12V_THIRD).
3. HARDWARE, above 12.0 A, about 150 us: U1321 ALERT latches SHED_N: third board, aux and spare off through 1k series FETs, no firmware. Firmware clears after the total stays under 10.5 A.
4. Above 12.4 A for 1 s with step 3 done: Pi stub and Jetson last, after telling the host. PG_12V falling alone never sheds compute: log only.
PORT SOFTWARE TRIP (firmware): each ADC_IMx, 1–15 A per port, default 15 A on arm and waist, 10 A on aux. Over the limit drops PORT_REQx, then reset needs the 148 ms latch sequence.
WATCHDOG: kick WD_KICK from the main loop only, after the port and pack checks ran. 1.36–1.84 s, not firmware-startable or stoppable.





Supervisor can open the banks and ports, never close them against the loop. All enables fail off.

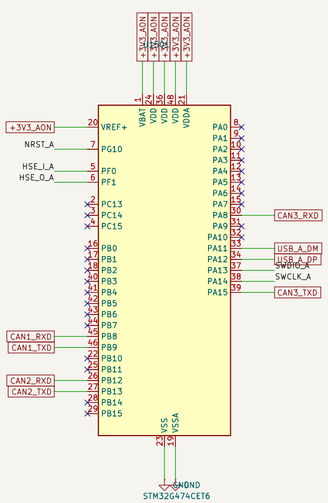
marengo

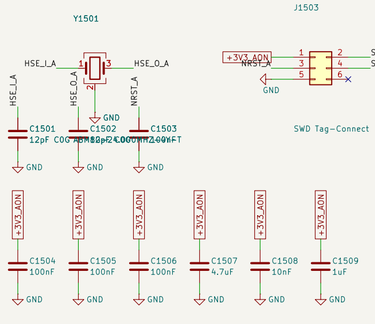
Sheet: /LP_logic_board/supervisor_mcu/
File: supervisor_mcu.kicad_sch

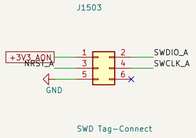
Title: supervisor_mcu: STM32G0B1 supervisor, I2C mux

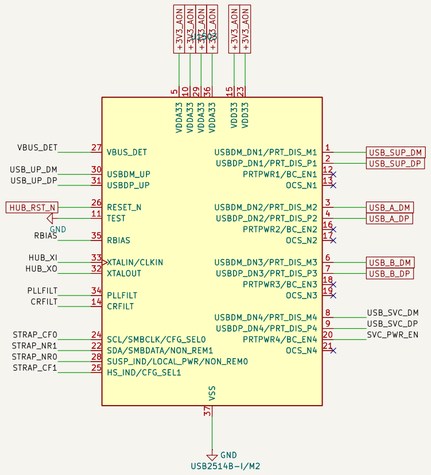
Size: A2	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 14/27

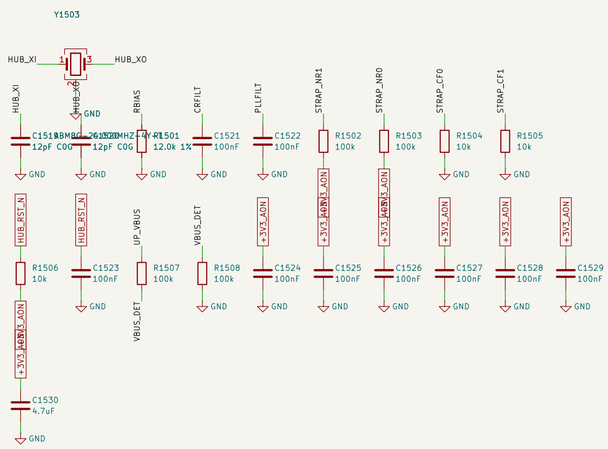
Hub: DN1 supervisor, DN2 bridge A, DN3 bridge B, DN4 service. No EEPROM: CFG_SEL 00, NON_REM 11, self powered, individual switching. TEST to GND. OCS_N1–4 open (internal pull-up).
FDCAN1_RX on PB8-BOOT0: program option bytes to ignore the pin before first run. PA15 released from JTDI for FDCAN3_TX. Firmware: budgetcan_fw, candleLight has no G474 build.
Service USB-C sits on a hub downstream port, so it is a host port: a laptop cannot reach the board through it. Rp 56k, default USB current.

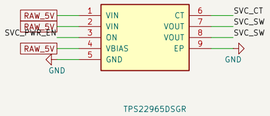


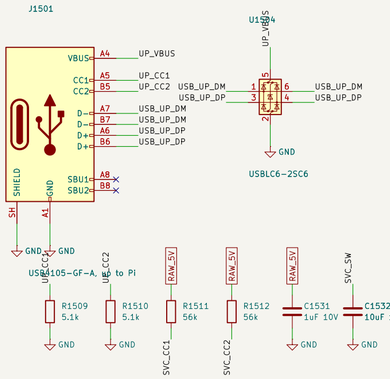


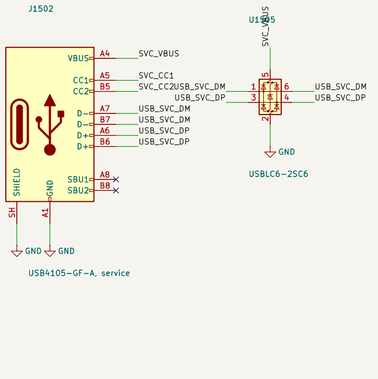












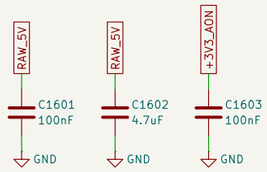
6 CAN buses over one USB-C to the Pi. Hub comes up without the supervisor, so the Pi keeps CAN if

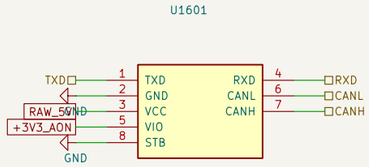
marengo

Sheet: /LP_logic_board/can_bridges_usb_hub/
File: can_bridges_usb_hub.kicad_sch

Title: can_bridges_usb_hub: two STM32G474 CAN bridges, USB2514B hub

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 15/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

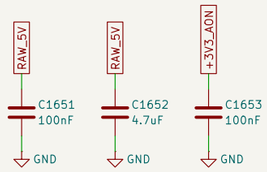
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

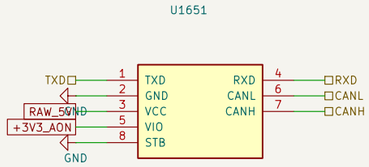
marengo

Sheet: /LP_logic_board/can_phy_1/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 16/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

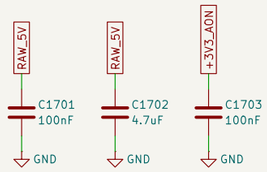
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

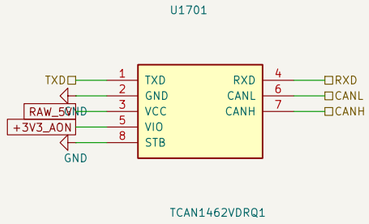
marengo

Sheet: /LP_logic_board/can_phy_2/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 17/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

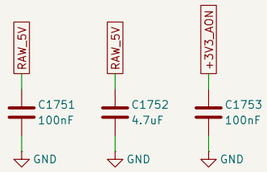
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

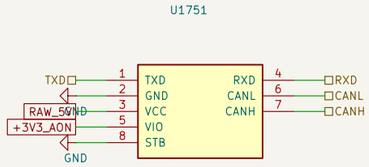
marengo

Sheet: /LP_logic_board/can_phy_3/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 18/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

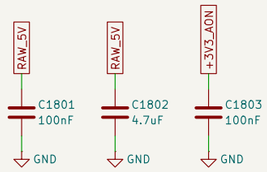
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

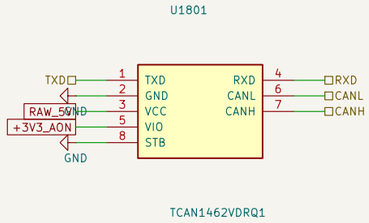
marengo

Sheet: /LP_logic_board/can_phy_4/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 19/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

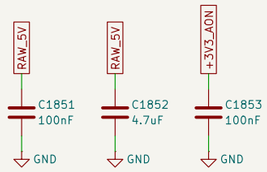
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

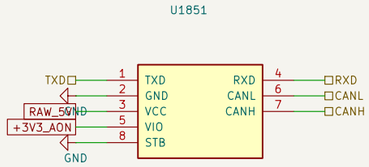
marengo

Sheet: /LP_logic_board/can_phy_5/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 20/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

marengo

Sheet: /LP_logic_board/can_phy_6/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 21/27

Ideal diode and hold-up. LM5050-1 not LM74700: PLAN 2.3 wants 75 V; LM74700 stops at 65 V, and VSYS needs no reverse-polarity function here.
20 x 68 uF 80 V hybrid, 10 x 10.8 mm = 1.36 mF, fits the 13 mm board gap: 12.6 ms at 90 W from 46.8 V, 6.8 ms from 37 V. README claimed 1 mF for 10 ms: true only above 48 V.
The 24 V converter taps VSYS ahead of the diode so 75 W does not drain the hold-up.

U1101
LM5050MK-1
Q1101
CSD19532Q5B
C1101 100nF 100V
C1102 100nF 100V
D1101 B340 30V
U1102
LM5145RGYR
Q1102
Q1103
BSC040N10NS5ATMA1
L1101 XAL1510-682MED 33.2k
R1101 33.2k
R1102 33.2k
R1103 2.37k
R1104 6.04k
C1123 10nF
C1124 33pF
R1105 100R
C1125 1nF
C1126 33nF
C1127 100nF 25V
R1106 2.2R
C1128 2.2uF 25V
R1107 348R
C1129 18pF C0G 100V
R1108 1.21M
R1109 49.9k
C1130 100nF 100V
R1110 20k
C1131 4.7uF 100V
C1132 4.7uF 100V
C1133 4.7uF 100V
C1134 4.7uF 100V
C1135 4.7uF 100V
C1136 22uF 25V
C1137 22uF 25V
C1138 22uF 25V
C1139 22uF 25V
C1140 22uF 25V
C1141 22uF 25V
C1142 22uF 25V
Q1104
BSS123
R1111 100k
U1103
LM5145RGYR
Q1105
Q1106
CSD19534Q5A
L1102 XAL1510-333MED 33.2k
R1112 33.2k
R1113 80.6k
R1114 2.8k
R1115 17.8k
C1143 6.8nF
C1144 10pF
R1116 100R
C1145 680pF
C1146 33nF
C1147 100nF 25V
R1117 2.2R
C1148 2.2uF 25V
R1118 301R
C1149 22pF C0G 100V
R1119 499k
R1120 18.2k
C1150 100nF 100V
R1121 20k
C1151 4.7uF 100V
C1152 4.7uF 100V
C1153 4.7uF 100V
C1154 10uF 50V
C1155 10uF 50V
C1156 10uF 50V
C1157 10uF 50V
C1158 10uF 50V
C1159 10uF 50V
C1160 10uF 50V
Hold-up bank, 20 x 68 uF, VSYS_OR to GND
C1103 68uF 80V B40950B0686M100
C1104 68uF 80V B40950B0686M100
C1105 68uF 80V B40950B0686M100
C1106 68uF 80V B40950B0686M100
C1107 68uF 80V B40950B0686M100
C1108 68uF 80V B40950B0686M100
C1109 68uF 80V B40950B0686M100
C1110 68uF 80V B40950B0686M100
C1111 68uF 80V B40950B0686M100
C1112 68uF 80V B40950B0686M100
C1113 68uF 80V B40950B0686M100
C1114 68uF 80V B40950B0686M100
C1115 68uF 80V B40950B0686M100
C1116 68uF 80V B40950B0686M100
C1117 68uF 80V B40950B0686M100
C1118 68uF 80V B40950B0686M100
C1119 68uF 80V B40950B0686M100
C1120 68uF 80V B40950B0686M100
C1121 68uF 80V B40950B0686M100
C1122 68uF 80V B40950B0686M100

BUCK12_OFF: pulled down, so a supervisor reset or hang leaves 12 V, compute and the E-stop loop running. Supervisor drives it high in Standby. BUCK24_OFF: pulled up, supervisor drives low to start 24 V.
The 12 V converter is the 300 kHz master; SYNCOUT drives the 24 V converter 180 degrees out of phase. 12 V feeds its own VCC through a Schottky after start-up (lower loss).

3.3 V always-on, on VSYS_OR after the ideal diode so it rides the hold-up bank with compute. EN tied to VIN. SNS- mode removes VSYS. 400 kHz, 3.3 V.

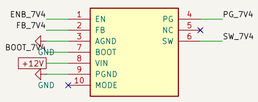
U1104
LMR38010SDDAR
L1103 XAL6060-223MEC 22uH
R1123 220k
R1124 43.2k
R1125 64.9k
C1161 4.7uF 100V
C1162 100nF 100V
C1163 100nF
C1164 22uF 25V
C1165 22uF 25V
C1166 22uF 25V
R1126 100k

LM5050-1, LM5145 x2, LMR38010. Hand design, values in RATIONALE-log.

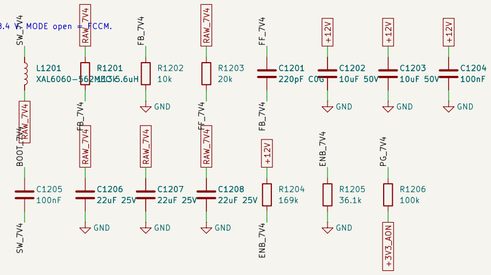
marengo		
Sheet: /LP_logic_board/front_end_12v_24v/		
File: front_end_12v_24v.kicad_sch		
Title: front_end_12v_24v: ideal diode, hold-up, 12 V, 24 V, 3.3 V always-on		
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 22/27

Servo rail. Adjust by R top: 90.9k = 6.05 V, 113k = 7.4 V, 130k = 8.4 V. MODE open = FCCM.

U1201

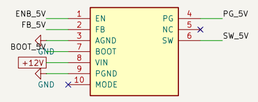


TPS56637RPAR

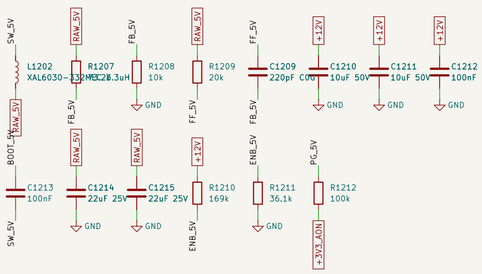


5 V peripheral, 4 A. EN divider per TI: starts near 10 V input.

U1202

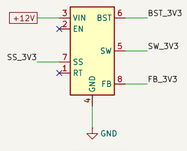


TPS56637RPAR

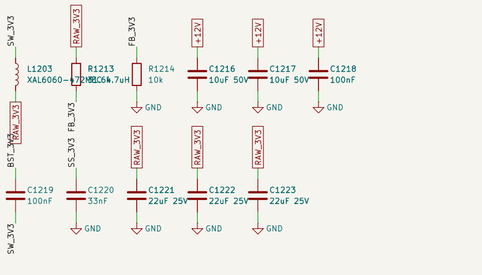


3.3 V peripheral, 2 A. RT open = 500 kHz, EN open = on.

U1203

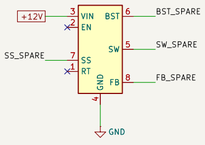


TPS62933DRLR

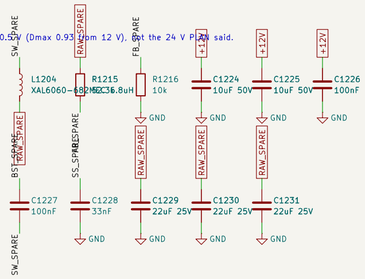


Spare rail, stuffed 5 V. 3.3 V: R top 31.6k. 9 V: see log. Max about 10.5 V (Dmax 0.93 from 12 V), but the 24 V PSU said.

U1204

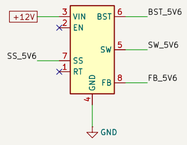


TPS62933DRLR

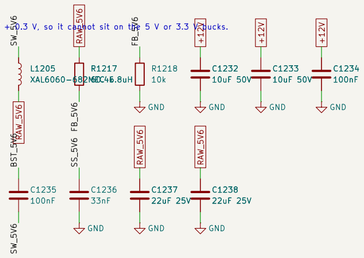


New: 5.6 V pre-regulator. TPS7A20 needs VIN <= 6.0 V and >= VOUT +0.3 V, so it cannot sit on the 5 V or 3.3 V bucks.

U1205

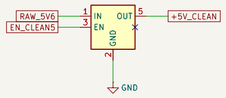


TPS62933FDRLR

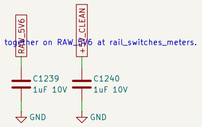


Clean rails, 0.3 A each, under 5 mV ripple. Metered together on RAW_5V6 at rail_switches_meters.

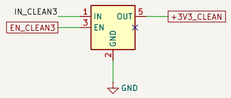
U1206



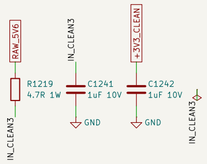
TPS7A2050PDBVR



U1207



TPS7A2033PDBVR





TPS56637 x2, TPS62933 x3, TPS7A20 x2. Hand design, values in RATIONALE-log.

marengo

Sheet: /LP_logic_board/pol_rails/

File: pol_rails.kicad_sch

Title: pol_rails: point-of-load rails from 12 V

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 23/27

OUT_24V: meter on bus A, straps A1,A0 = GND,GND, shunt 10mR

U1305
TPS26631PWPR
R1301 4.99k
R1302 475k
R1303 7.87k
R1304 22.1k
C1301 22nF
C1302 1uF 50V
C1303 100nF 50V
C1304 100nF 50V
R1305 33k
R1306 10mR 3W
R1307 10R
R1308 10R
C1305 100nF
C1306 100nF
U1306
INA238AIDGSR
OUT_24V
SP_24V
SN_24V
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

+12V total: converter output to rail, bus A, straps VS,VS.
ALERT at 12.0 A, latched, 150 us conversion, sheds third board, aux, spare in hardware. Firmware clears the latch.

R1368 2mR 3W
R1369 10R
R1370 10R
C1355 100nF
C1356 100nF
R1371 10k
SHED_N

OUT_12V_JETSON: meter on bus A, straps A1,A0 = GND,VS, shunt 10mR

U1303
TPS26631PWPR
R1309 6.04k
R1310 442k
R1311 21.5k
R1312 41.2k
C1307 47nF
C1308 1uF 50V
C1309 100nF 50V
C1310 100nF 50V
R1313 33k
R1314 10mR 3W
R1315 10R
R1316 10R
C1311 100nF
C1312 100nF
U1304
INA238AIDGSR
OUT_12V_JETSON
SP_12V_JETSON
SN_12V_JETSON
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

OUT_12V_THIRD: meter on bus A, straps A1,A0 = GND,SDA, shunt 10mR

U1305
TPS26631PWPR
R1317 6.04k
R1318 442k
R1319 21.5k
R1320 41.2k
C1313 47nF
C1314 1uF 50V
C1315 100nF 50V
C1316 100nF 50V
R1321 33k
R1322 10mR 3W
R1323 10R
R1324 10R
C1317 100nF
C1318 100nF
U1307
INA238AIDGSR
OUT_12V_THIRD
SP_12V_THIRD
SN_12V_THIRD
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

Q1301 2N7002
Q1302 2N7002
U1322
SN74LVC1G14DBV
SHED_12V
SHED_N

OUT_12V_AUX: meter on bus A, straps A1,A0 = GND,SCL, shunt 10mR

U1302
TPS26631PWPR
R1325 4.53k
R1326 442k
R1327 21.5k
R1328 41.2k
C1319 47nF
C1320 1uF 50V
C1321 100nF 50V
C1322 100nF 50V
R1329 33k
R1330 10mR 3W
R1331 10R
R1332 10R
C1323 100nF
C1324 100nF
U1308
INA238AIDGSR
OUT_12V_AUX
SP_12V_AUX
SN_12V_AUX
I2C_A_SCL
I2C_A_SDA
RAIL_ALERT_A

OUT_12V_PI_STUB: meter on bus A, straps A1,A0 = VS,GND, shunt 10mR

U1309
TPS26631PWPR
R1333 4.99k
R1334 442k
R1335 21.5k
R1336 41.2k
C1325 47nF
C1326 1uF 50V
C1327 100nF 50V
C1328 100nF 50V
R1337 33k
R1338 10mR 3W
R1339 10R
R1340 10R
C1329 100nF
C1330 100nF
U1310
INA238AIDGSR
OUT_12V_PI_STUB
SP_12V_PI
SN_12V_PI
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

OUT_7V4_SERVO: meter on bus B, straps A1,A0 = GND,GND, shunt 5mR

U1311
TPS26631PWPR
R1341 3.01k
R1342 383k
R1343 64.9k
R1344 64.9k
C1331 47nF
C1332 1uF 50V
C1333 100nF 50V
C1334 100nF 50V
R1345 33k
R1346 5mR 3W
R1347 10R
R1348 10R
C1335 100nF
C1336 100nF
U1312
INA238AIDGSR
OUT_7V4_SERVO
SP_7V4
SN_7V4
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_5V_PERIPH: meter on bus B, straps A1,A0 = GND,VS, shunt 5mR

U1313
TPS22965DSGR
F1301 2.5A MF-MSMF250/16X
R1350 10mR 3W
R1351 10R
R1352 10R
C1337 1uF 10V
C1338 1nF
C1339 10uF 10V
R1349 100k
C1340 100nF
C1341 100nF
U1314
INA238AIDGSR
OUT_5V_PERIPH
SP_5V
SN_5V
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_3V3_PERIPH: meter on bus B, straps A1,A0 = GND,SDA, shunt 10mR

U1315
TPS22965DSGR
F1302 1.1A MF-MSMF110/16X
R1354 10mR 3W
R1355 10R
R1356 10R
C1342 1uF 10V
C1343 1nF
C1344 10uF 10V
R1353 100k
C1345 100nF
C1346 100nF
U1316
INA238AIDGSR
OUT_3V3_PERIPH
SP_3V3
SN_3V3
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_SPARE: meter on bus B, straps A1,A0 = GND,SCL, shunt 10mR

U1317
TPS26631PWPR
R1357 4.99k
R1358 365k
R1359 33.2k
R1360 97.6k
C1347 47nF
C1348 1uF 50V
C1349 100nF 50V
C1350 100nF 50V
R1361 33k
R1362 10mR 3W
R1363 10R
R1364 10R
C1351 100nF
C1352 100nF
U1318
INA238AIDGSR
OUT_SPARE
SP_SPARE
SN_SPARE
I2C_B_SCL
I2C_B_SDA
RAIL_ALERT_B

RAW_5V6_M: meter on bus B, straps A1,A0 = VS,GND, shunt 10mR

U1319
TPS26631PWPR
R1365 10mR 3W
R1366 10R
R1367 10R
C1353 100nF
C1354 100nF
U1320
INA238AIDGSR
RAW_5V6_M
SP_CLEAN
SN_CLEAN
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

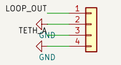
TPS26631 OV/OV chain mandatory (UVLO to GND means 15.46 V, would kill a 12 V rail). Latch mode, supervisor clears. SHDN default off.
TPS22965 has no current limit. Its 6 A is under the buck limit; polyfuse after each switch. 5 V 2.5 A, 3.3 V 1.1 A.
Output limits: 24 V 3.61 A, Jetson 2.98 A, third 2.98 A, aux 3.97 A, PI stub 3.61 A, 7.4 V 5.98 A, spare 3.61 A (TPS26631 latch off), 5 V 2.5 A, 3.3 V 1.2 A (polyfuse, hold at 23 C, derates hot).
INA238 has no energy registers: firmware integrates at 100 Hz, or swap to INA228 (same footprint).

R1372 100k
R1373 10k
R1374 10k

TPS26631 eFuse / TPS22965 switch + INA238 per output. 2 I2C buses, 11 meters. Hardware 12 V sh	
marengo	
Sheet: /LP_logic_board/rail_switches_meters/	
File: rail_switches_meters.kicad_sch	
Title: rail_switches_meters: rail switches and meters	
Size: A1 | Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8	Id: 24/27

E-stop loop, 6.9 mA: +12V, 680R, NC contacts, brake thermostat, opto A, opto B, 620R, GND. No +12 V on the loop connectors: a pin-to-pin short lands on GND and opens the loop.

J2001



E-stop loop

J2002



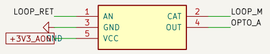
tether, shorting dummy

J2003



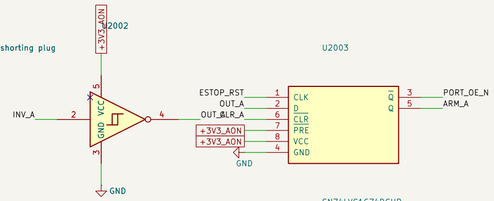
wireless contact, shorting plug

U2001



ISOM8710

U2002



SN74LVC1G74DCUR

U2003

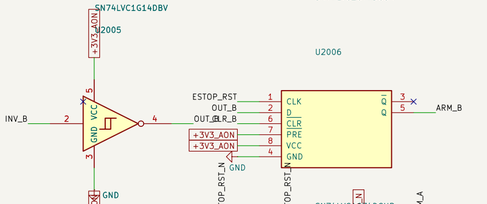
SN74LVC1G14DBV

U2004



ISOM8710

U2005



SN74LVC1G74DCUR

U2006

SN74LVC1G14DBV

J2004



brake thermostat NC

J2005



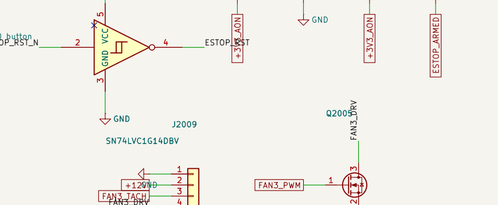
wireless receiver power

J2006



E-stop reset, NO button

U2007



SN74LVC1G14DBV

J2009



fan 1

Q2003



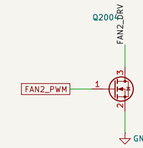
2N7002

J2008

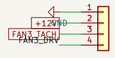


fan 2

Q2004



2N7002



fan 3

Q2005

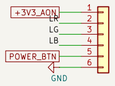


2N7002

Fans, 12 V, 25 kHz PWM open drain (polarity inverts in firmware).

User I/O: 6-pin shell cable, the plan's 4 pins cannot carry RGB + button. Bench jumper, buzzer, charge LED, Qwiic, air NTC.

J2010



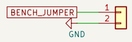
shell: button + RGB

J2011



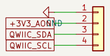
charge LED

J2012



bench jumper

J2013



Qwiic

J2014



intake NTC

J2015



exhaust NTC

Rail outputs, one Micro-Fit each. TS2 wake: button low pulls AFE TS2 to VSS through 10k + BAT54 with the board off.

J2016



OUT_24V

J2017



OUT_12V_JETSON

J2018



OUT_12V_THIRD

J2019



OUT_12V_AUX

J2020



OUT_12V_PI_STUB

J2021



OUT_7V4_SERVO

J2022



OUT_5V_PERIPH

J2023



OUT_3V3_PERIPH

J2024



OUT_SPARE

J2025

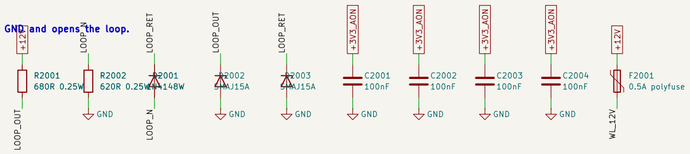


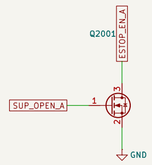
5V_CLEAN

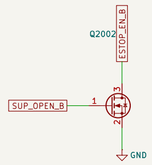
J2026

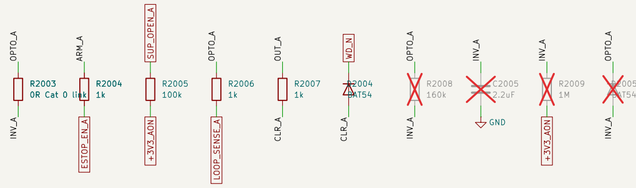


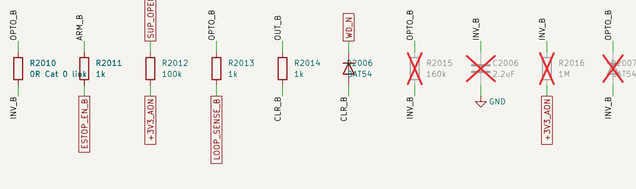
3V3_CLEAN



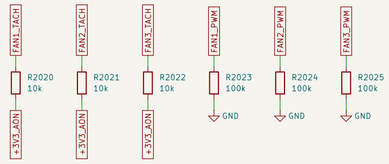




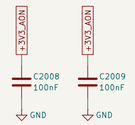




SUP_OPEN high: 2N7002 sinks EN. MCU dead or Hi-Z: pull-up opens the banks. The FET only sinks, so firmware cannot close an open loop.
Cat 1 option: remove the OR, fit the four DNP parts, 289 ms nominal (232-356 ms). DNP in rev A: Cat 0.
Re-close latch: loop open, watchdog or power-up clears the flops. Re-arming needs a rising edge from the NO reset button with the loop closed. A held button does nothing. No firmware path.
One series loop still feeds both channels: a short across one contact pair defeats both. Not Cat 3.

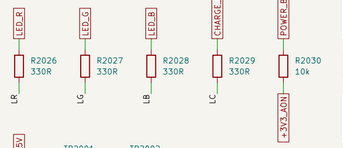


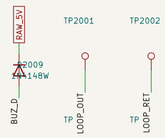
Hardware watchdog and port gate. Starts itself at power-up, no firmware can start or stop it:
SET1 high, CWD open = 1.6 s (1.36-1.84 s). Timeout pulls WD_N low, clears both E-stop latches.
PORT_OE_N follows latch A, so every PORT_EN tri-states and its 100k pulls it off. Compute rails untouched.



U2031

TPS3431SDRBR

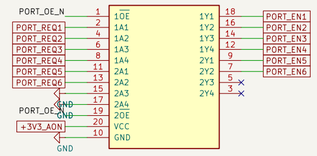








U2032



SN74LVC244APWR

E-stop path: loop current to opto to inverter to gate-driver enable. No firmware in that path.

marengo

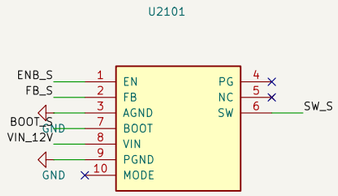
Sheet: /LP_logic_board/io_fans_estop_loop/
File: io_fans_estop_loop.kicad_sch

Title: io_fans_estop_loop: E-stop loop, fans, user I/O, output connectors

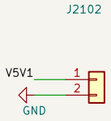
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 25/27



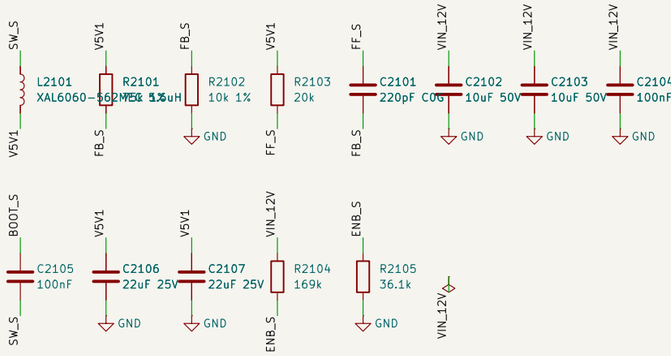
12 V in



TPS56637RPAR



5.1 V out, 8.5 A per contact



Output pigtail goes to a USB-C plug. A 2x3 housing on the GPIO header would put all 6 A through one GND pin (pin 6) and cannot lock.
Set usb_max_current_enable=1 on the Pi. PG left open: no logic rail on the stub.

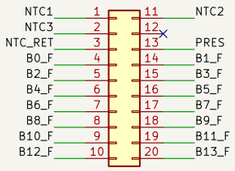
Own 2-layer board. Local nets; harness map checked in check.py.

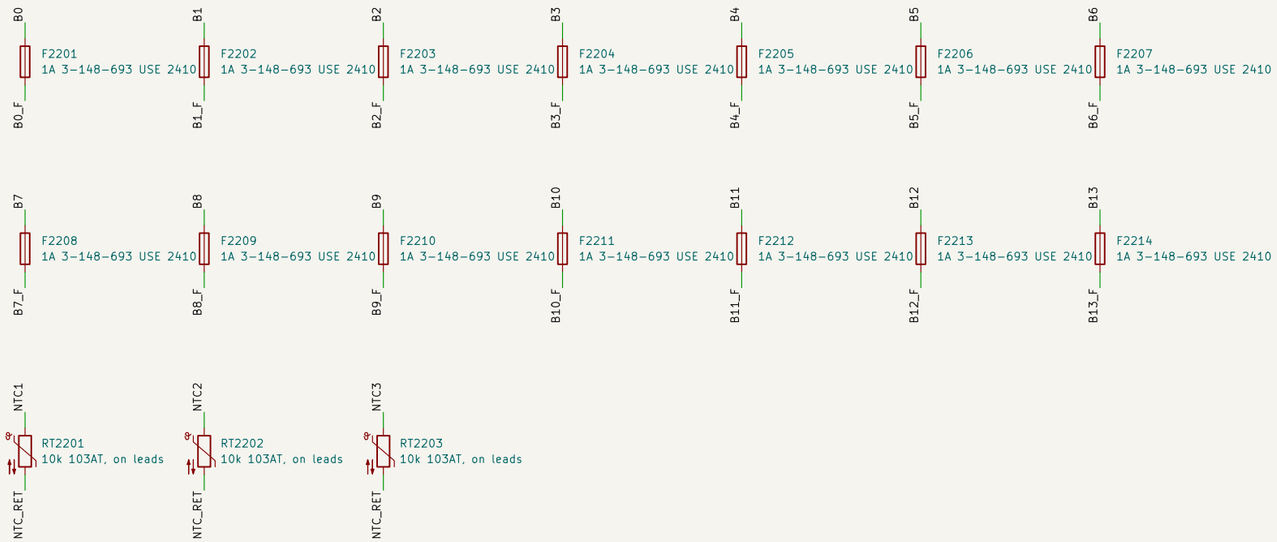
marengo

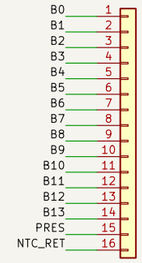
Sheet: /pi_power_stub/
File: pi_power_stub.kicad_sch

Title: pi_power_stub: 12 V to 5.1 V 6 A at the Pi

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 26/27







Fuse at the cell, one per tap. Schurter USE 2410 1 A: 79 mOhm, 200 A at 86 VDC, tau under 0.05 ms, against 146 A prospective. Balance adds 24 mV of sense error at 150 mA across two taps.
Footprint is a 2010 stand-in: the 6.1 x 2.5 mm 2410 land is owed in layout.
Presence: pin 13 goes out through the cell tap plug (J2202 pins 15, 16 looped in the harness) and back on pin 3, so either plug out opens DSG and CHG. B0 and B13 are their own Kelvin tabs: no pack current in a sense wire. NTC_RET returns on pin 3 to VSS, no cell connection.

Own 2-layer board at the pack. Pin map must match HP J2. Local nets; harness map checked in che

marengo

Sheet: /pack_sense_board/
File: pack_sense_board.kicad_sch

Title: pack_sense_board: tap fuses, NTC, presence loop

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 27/27